

How to create a Switch

// ——— IMPORTS (top of file, before everything)

// every widget/class you use needs an import up here

import android.os.Bundle // required for onCreate()

import android.widget.Button // for Button widget

import android.widget.Switch // for Switch toggle widget

import android.widget.Toast // for Toast popup messages

import androidx.appcompat.app.AppCompatActivity // base class all activities extend

// ——— CLASS (one class = one screen)

class MainActivity : AppCompatActivity() {

// ——— onCreate() — runs once when screen opens

// ALL your logic lives inside this function

override fun onCreate(savedInstanceState: Bundle?) {

super.onCreate(savedInstanceState) // always the first line — sets up the activity

setContentView(R.layout.activity_main) // loads activity_main.xml onto the screen

// — 1. TOAST on button click

// connect to your Button in XML using its ID

val btnToast = findViewById<Button>(R.id.btnToast) // R.id must match
android:id in XML

btnToast.setOnClickListener { // runs when the button is tapped

Toast.makeText(

```

    this,          // context = current screen
    "Hello Mpho!", // message shown in popup
    Toast.LENGTH_SHORT // SHORT = 2s, LONG = 3.5s
).show()          // .show() is REQUIRED – without it nothing appears
}

```

// — 2. SWITCH toggle

```

// connect to your Switch widget in XML using its ID
val mySwitch = findViewById<Switch>(R.id.mySwitch) // R.id must match
android:id in XML

```

```

// fires every time the user flips the switch
// _ = the switch view (ignored), isChecked = true if ON / false if OFF
mySwitch.setOnCheckedChangeListener { _, isChecked ->
    if (isChecked) { // switch flipped ON
        Toast.makeText(this, "Switch is ON", Toast.LENGTH_SHORT).show()
    } else {         // switch flipped OFF
        Toast.makeText(this, "Switch is OFF", Toast.LENGTH_SHORT).show()
    }
}
}

```

// — 3. WHEN – match a string value

```

val day = "Monday" // value to check

when (day) { // Kotlin's switch – checks value of day top to bottom
    "Monday" -> Toast.makeText(this, "Start of the week",
Toast.LENGTH_SHORT).show()
}

```

```
        "Friday" -> Toast.makeText(this, "Almost weekend!",  
Toast.LENGTH_SHORT).show()
```

```
        "Sunday" -> Toast.makeText(this, "Rest day",  
Toast.LENGTH_SHORT).show()
```

```
        else -> Toast.makeText(this, "Normal day",  
Toast.LENGTH_SHORT).show()
```

```
        // else = default, runs if nothing above matched
```

```
    }
```

```
    // — 4. WHEN — number ranges + expression
```

```
        val score = 85 // number to check
```

```
        // when as expression — returns a value directly into grade
```

```
        val grade = when (score) {
```

```
            in 90..100 -> "A" // in = checks if score is inside this range
```

```
            in 75..89 -> "B" // score 85 lands here so grade = "B"
```

```
            in 60..74 -> "C"
```

```
            else -> "F" // default fallback
```

```
        }
```

```
        // $grade = string interpolation, swaps $grade for its actual value
```

```
        // toast shows: "Your grade: B"
```

```
        Toast.makeText(this, "Your grade: $grade", Toast.LENGTH_SHORT).show()
```

```
    } // ← end of onCreate() — nothing goes outside this except the class closing brace
```

```
} // ← end of MainActivity class
```